PyMOL Tutorial

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上海科技大学

研究方向：蛋白质设计；蛋白质工程；结构生物学

- 上海市海外高层次引进人才
- 浦江人才
- 国家重点研发计划合成生物学重点专项
- 国家重点研发计划生物大分子与微生物组重点专项
- 国家自然科学基金委青年科学基金项目

Education & Research Background

- 2003 - 2007 **北京大学**

  生物科学　　学士



- 2007 - 2012 **University of Washington**

  结构生物学　博士　　导师: 许文青教授



- 2013 - 2019 **University of Washington**

  博士后　　导师: 许文青 / David Baker 教授



- 2019 - 至今　**上海科技大学**　　生命科学与技术学院

  副研究员



Outline

- Part I: Introduction
- Part II: Case study

PyMOL:

a structural visualization and analysis program





Warren L. DeLano, (1972-2009) BS. Yale (1993); PhD, UCSF 1999.

Pymol maintained and distributed by Schrödinger (https://pymol.org/)

Running on Windows, Mac OS, Linux, …

Pymol user guide:
pymol.sourceforge.net/newman/userman.pdf

Practical Pymol for Beginners:
https://pymolwiki.org/index.php/Practical_Pymol_for_Beginners

Download PyMOL 3.0

Version 3.0.5 - Updated October 8th 2024 (Installation instructions)

For previous versions see here.

These bundles include Python 3.10.

Pymol 2.5.7

Windows	Windows	macOS	Linux
EXE Installer	ZIP Archive	DMG Disk Image	TAR.BZ2 Archive

Or install from the Schrodinger Anaconda Channel.

```
conda install -c conda-forge -c schrodinger pymol-bundle
```

New users:

BUY LICENSE

Existing users:

DOWNLOAD LICENSE FILE

Buy PyMOL

Order PyMOL and AxPyMOL Online

Please select your intended use from the list below to receive access within one business day:

- Commercial

  for corporate organizations

- Government

  for government institutions

- Academic/Non-profit

  for academic and non-profit researchers/educators

- Student/Teacher

  Educational subscriptions to PyMOL are available at no cost to full-time students and educators teaching full-time students. However, there is not an educational subscription for AxPyMOL at this time. Under this agreement, Schrödinger has no obligation to furnish updates, fix bugs, provide documentation, and meet other support needs. Please recognize that long-term reliance upon PyMOL for educational purposes should merit a subscription purchase in order to secure essential services and baseline accountability.

All paid subscriptions include software upgrades for the duration of the subscription term, technical support in English, and access to Official Online Documentation.

How to use pymol…

1. loading pdb files (PDB: http://www.wwpdb.org, https://alphafold.ebi.ac.uk/)

2. five buttons (right panel) : action, show, hide, label, color 

3. mouse mode (right bottom panel) : rotate, zoom, slap, select atoms or residues…

Useful techniques..

1. changing background : display – background – select your favorite color

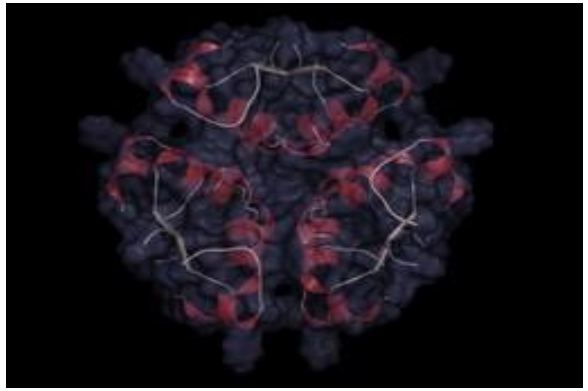
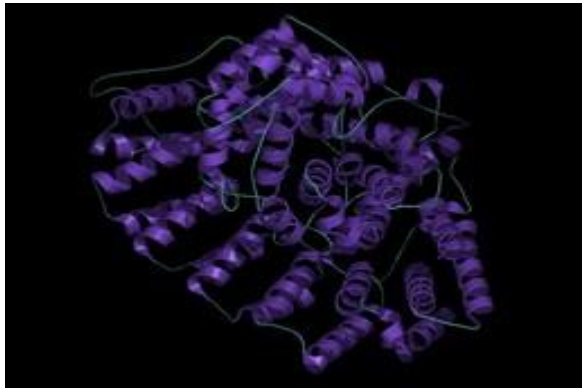
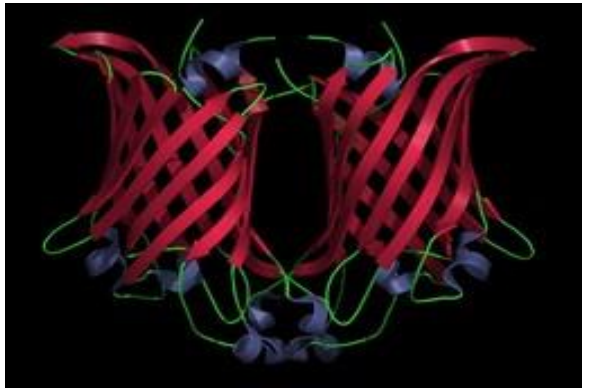
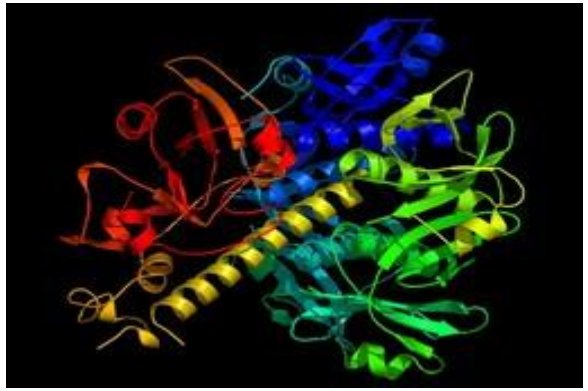
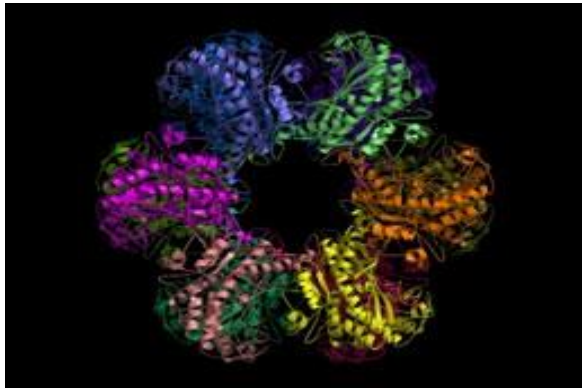
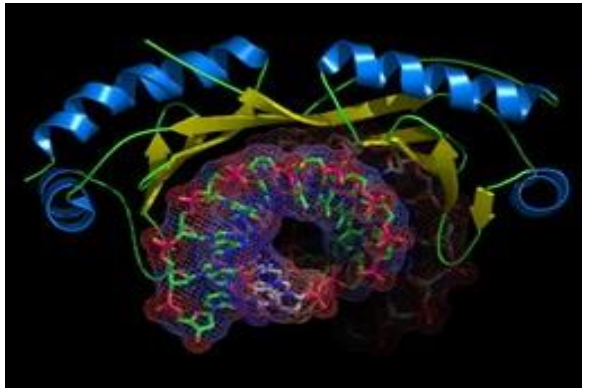
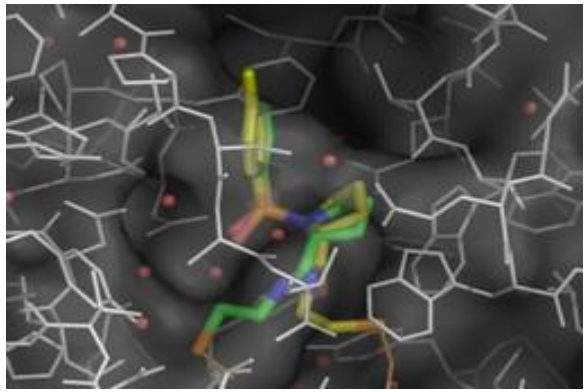
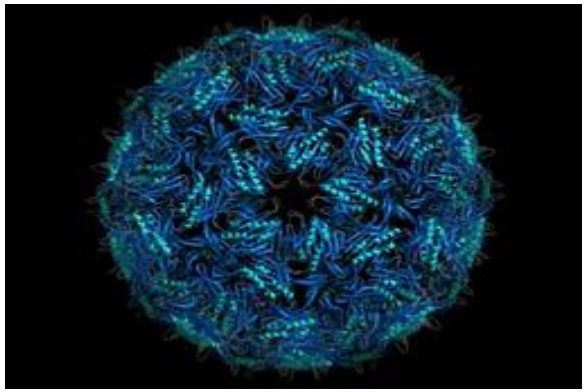
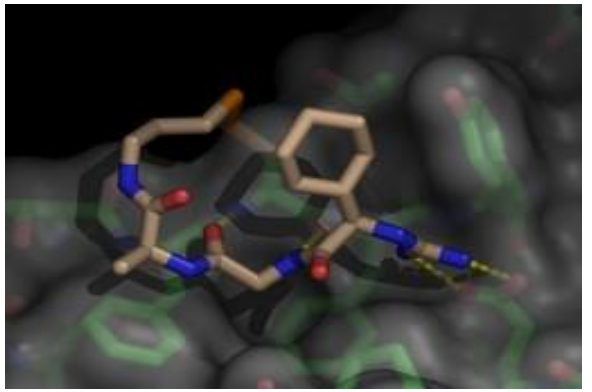
2. selection
   command : select object-name, selection-expression

3. show sequence, and select from the sequence

4. generate molecule surface : right panel – show - surface

5. transparency : setting – transparency – select cartoon/surface/….

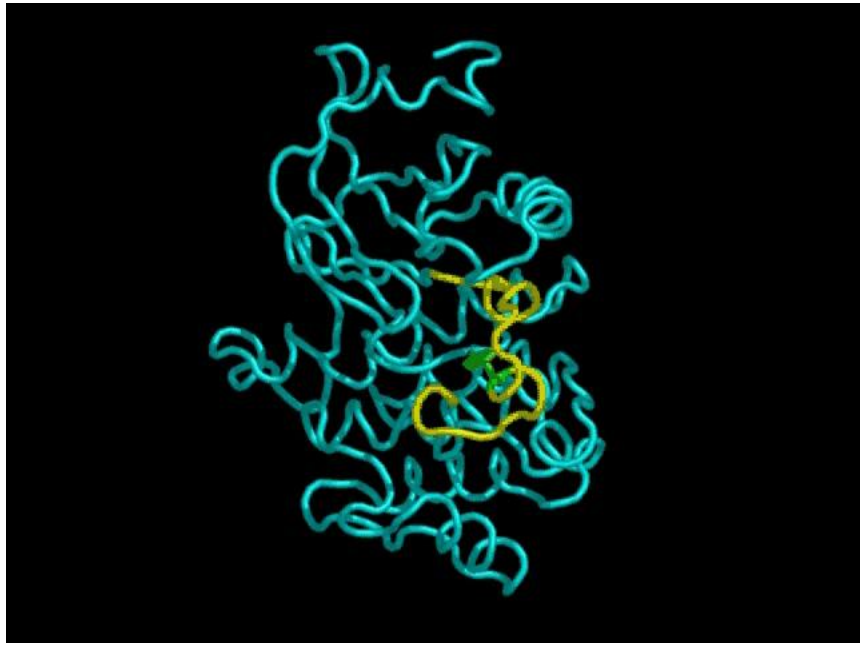
6. distance measuring : wizard - distance

More advanced work …

1. making movies
2. alignment of two similar protein structures
3. morph

Src kinase structure

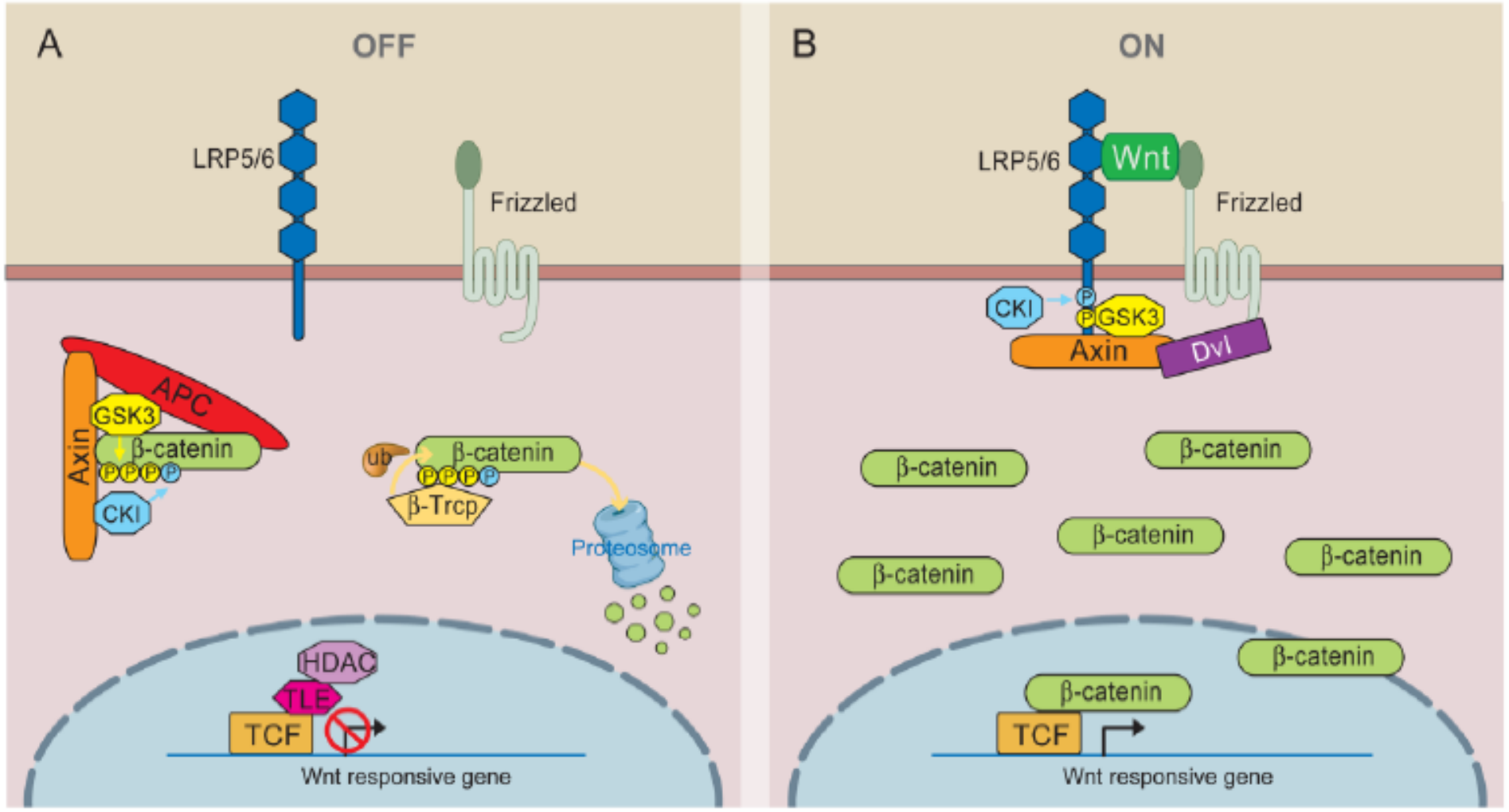


Xu, *et al.* Nature (1997)

Case study:

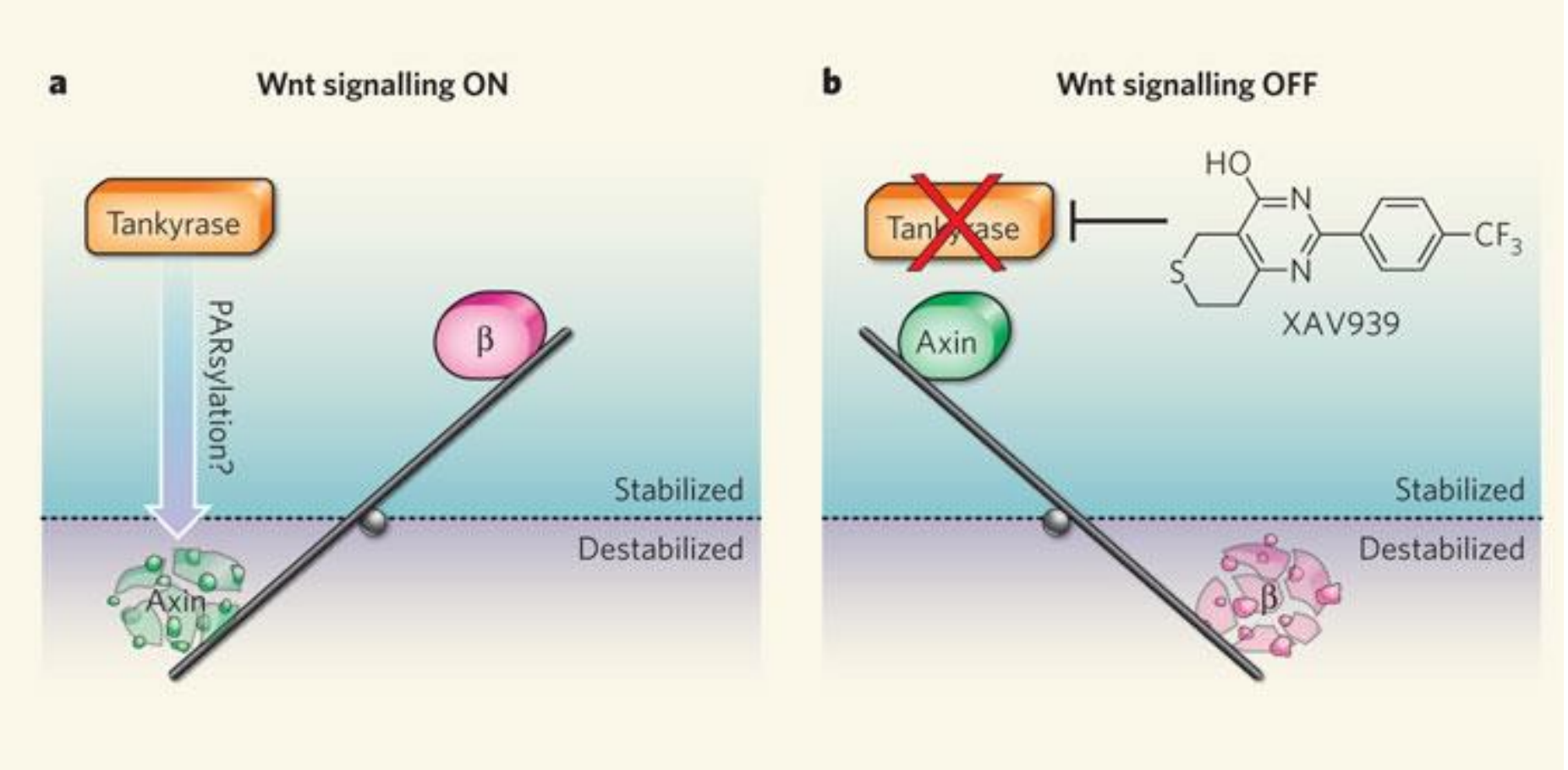
iso-ADPr / RNF146 structures

Canonical Wnt-signaling pathway



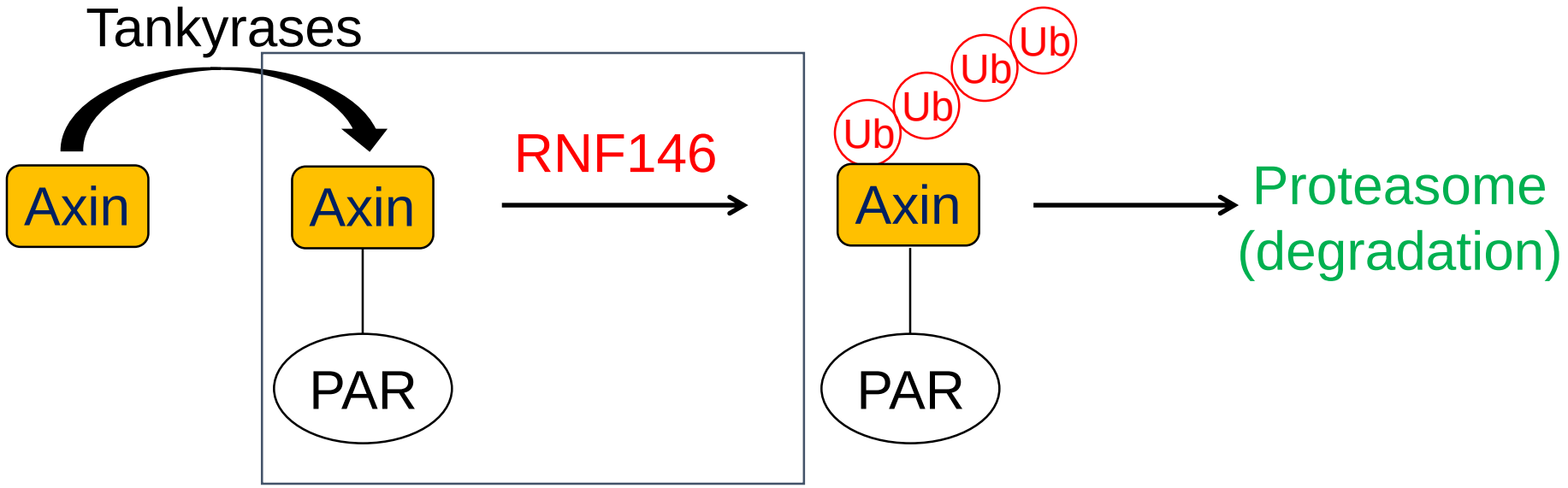
MacDonald *et al. Dev Cell.* (2009)

Axin turnover is controlled by poly(ADP-ribosyl)ation



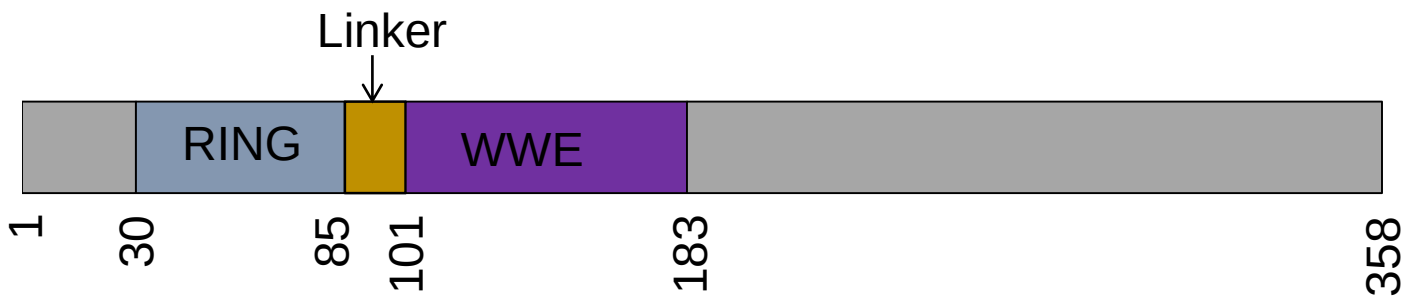
Huang *et al. Nature.* (2009)
R.T. Peterson *Nature* (2009)

RNF146 catalyzes poly(ADP-ribosyl)ation dependent Poly-ubiquitination of Axin

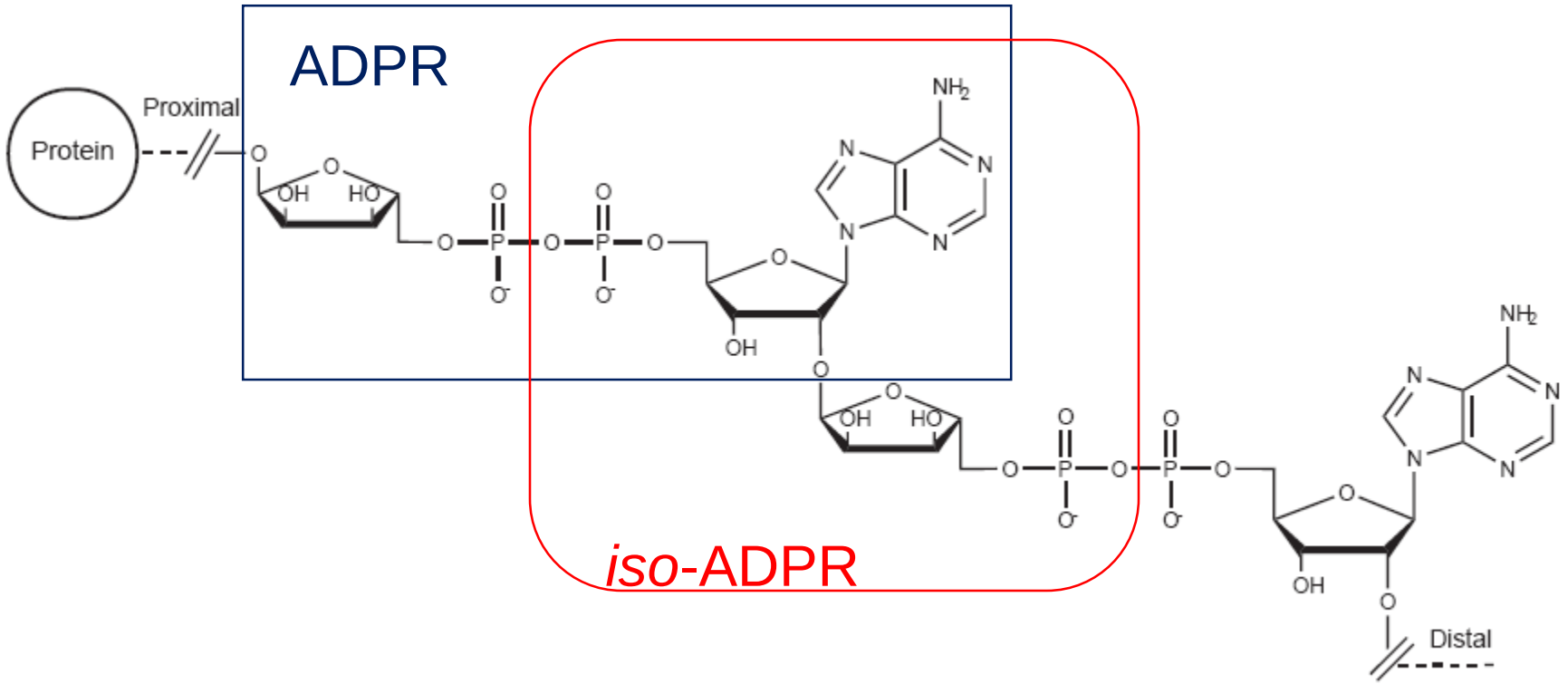


Zhang *et al. Nat. Cell Biol.* (2011)
Kang et al. *PNAS USA* (2011)

RNF146 as a RING E3 ligase

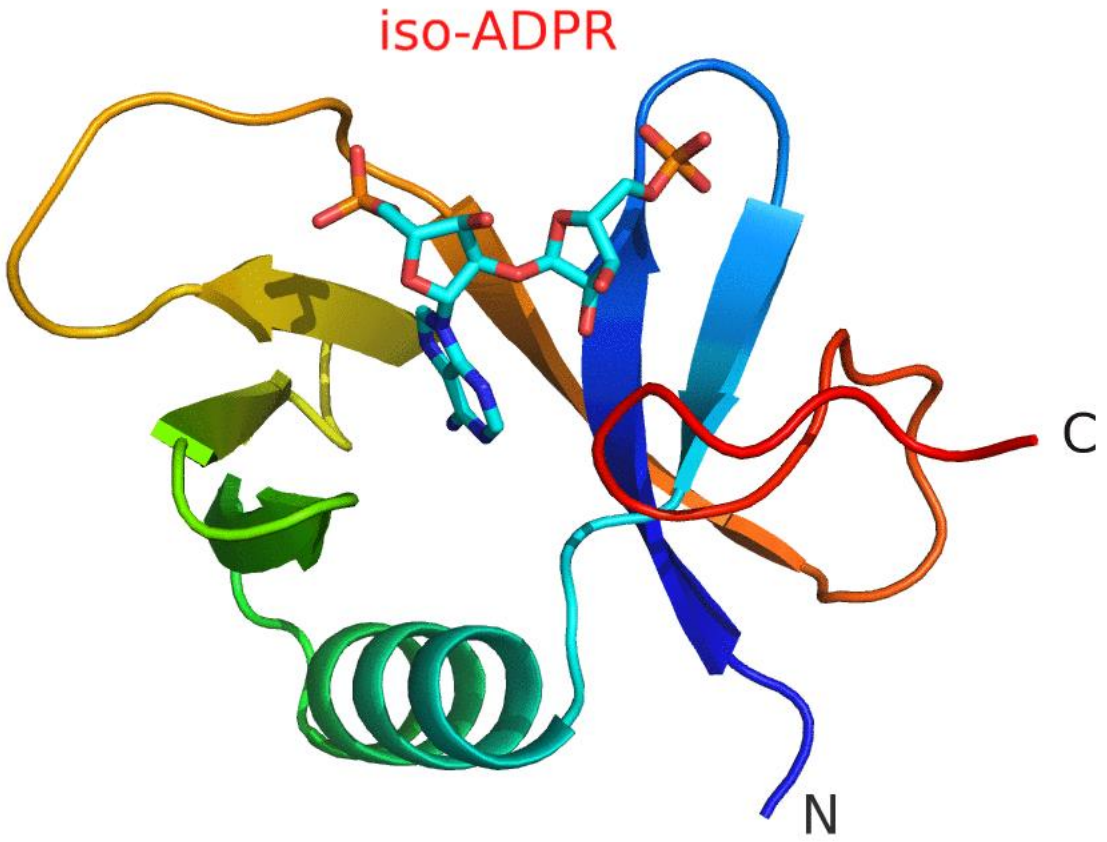


PAR polymer and its structural units



Wang *et al. Genes Dev.* (2012)

Overall Structure

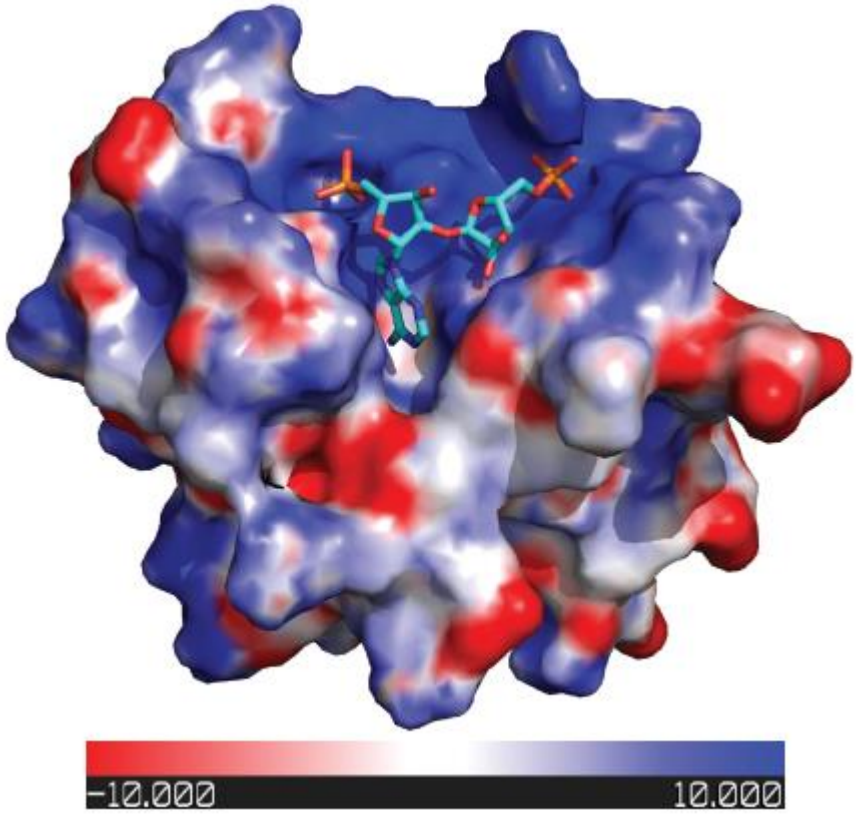


1.6 Å resolution

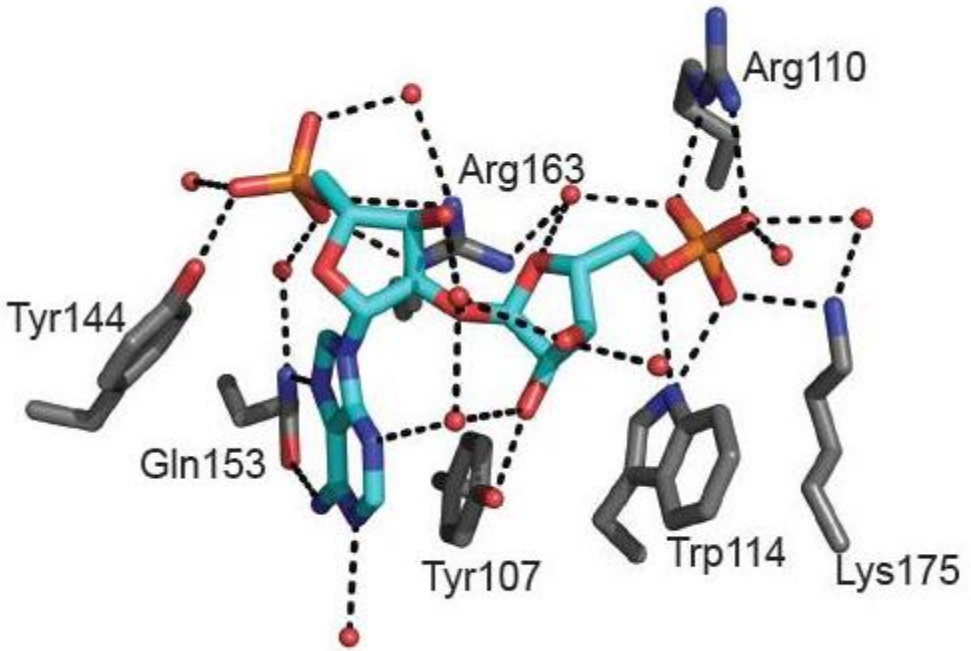
PDB: 3V3L

Wang *et al. Genes Dev.* (2012)

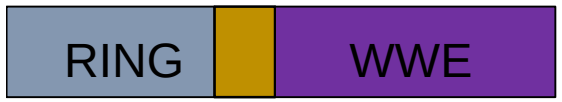
Electrostatic Surface Potentials

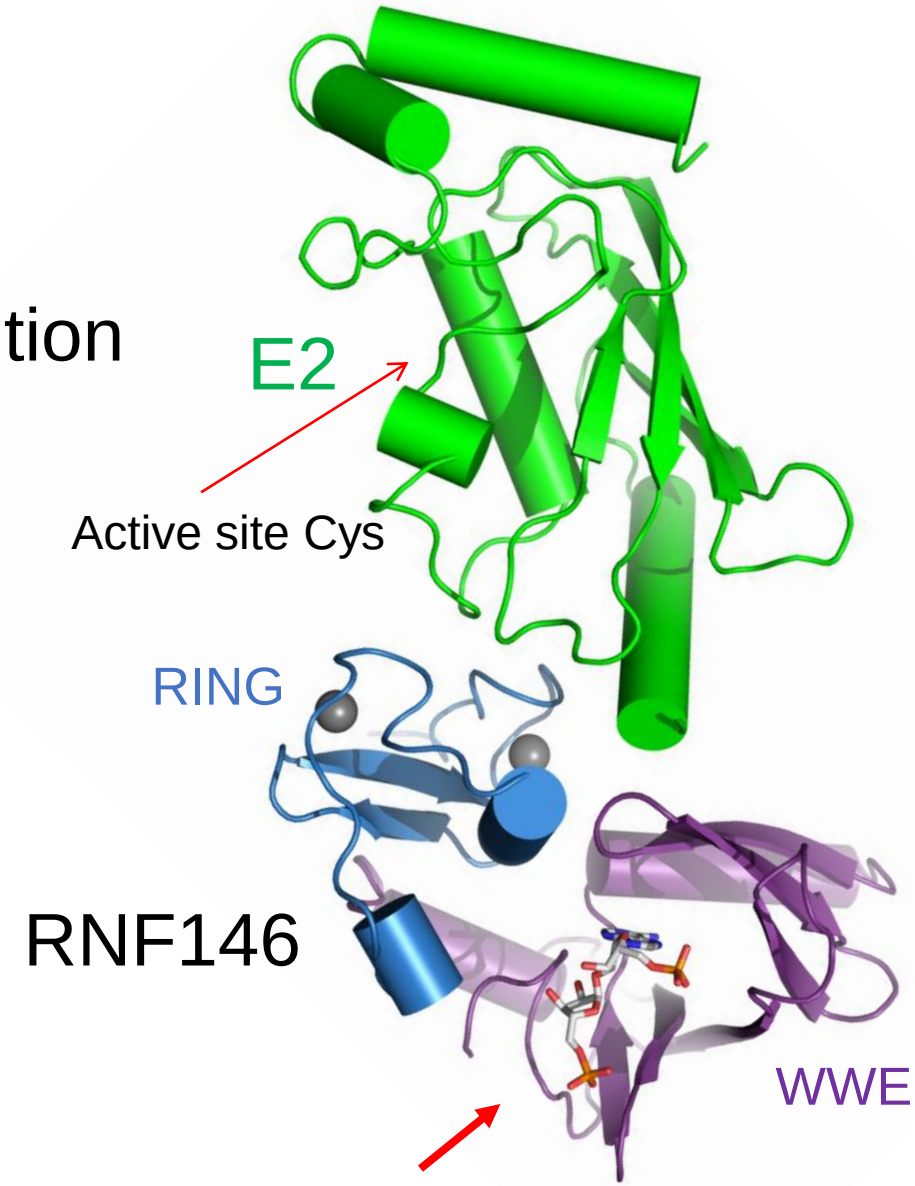


Ligand Binding



Crystal Structure of the E2/RNF146/*iso*-ADPR complex at 1.9 Å resolution





PDB: 4QPL

DaRosa & Wang, *et al. Nature* (2015)

Thanks!